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The renewable glassy carbon annular band electrode in a highly sensitive normal pulse voltammetric determination of paracetamol with continuous wavelet transformation

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ABSTRACT

This work describes a novel working electrode – the renewable glassy carbon annular band electrode (R-GC-ABE). The strategy to create the R-GC-ABE is based on the mechanical removal of solid and gas contaminants and electrochemical activation in the electrolyte which fills the electrode body. The main constituents of the R-GC-ABE: a specially constructed glassy carbon annular band electrode, a silver wire used as silver counter/quasi-reference electrode and a silicon O-ring are fastened together in a polypropylene body.

The R-GC-ABE was constructed and evaluated as a sensor for determination of paracetamol (PAR) by differential pulse (DP) and normal pulse (NP) voltammetry. Also a signal processing algorithm which utilizes continuous wavelet transform was described and applied for transformation of the sigmoidal-shaped NPV signals – to the peak-shaped curves.

In the effect, the new strategy of PAR determination is 32 times more sensitive than DPV technique. The repeatability is below 2–3% ($n = 3$) for the 0–50 mg L^{-1} concentration range of PAR. The detection limit is 4 μg L^{-1} with sensitivity of 2.75 ± 0.04 μA/mg L^{-1}. The correlation coefficient is equal to 0.9990.

The proposed method was successfully applied for determination of PAR in commercial tablets and in spiked human urine (CRM). The obtained results show substantial improvement of the performance of R-GC-ABE electrode and NPV technique.

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1. Introduction

Carbon has been a widely used as electrode material since Michael Faraday, and has been extensively studied for use as working electrode, both with and without modification [1,2]. The period of wide applications of carbon electrodes (CEs) in electroanalysis started with the invention of the carbon-paste electrode (CPE) by Adams in 1958, followed by wax impregnated graphite and then pyrolytic graphite and glassy carbon (GC), both introduced in analytical voltammetry by Miller and Zittel in 1963 and 1965 [3]. CEs are chemically inert, have a rich surface chemistry and are suitable for determination of a variety of different substances, especially in the anodic potential area [4]. Carbon based electrodes are fabricated from heterogeneous or homogenous materials, and the most popular forms are glassy carbon, carbon paste, carbon fiber, graphite, screen-printed carbon strips, carbon films or other composites [4–7]. The progress in modern engineering has led to development of new carbon based materials with assigned properties including different graphites, porous carbon, nanotubes, microelectrodes or boron doped diamonds [8–13].

Some of commonly met problems with the use of GCEs are degradation of the surface, distortion of peak shape and the emergence of additional peaks [4,5]. These may be caused by the adsorption of surface – active substances, organic matrix etc. on the GCEs surface and appearance of the redox processes leading to the nucleation or deposition of products layer. As a consequence, sensitivity and selectivity of the GCEs deteriorate. Different pretreatments procedures for preparing and activating the GC electrode surface for electrochemical measurements have been widely discussed [1,2,5]. Fresh and well-defined electrode surfaces can by prepared by mechanical treatment. Other methods, such as laser treatment, irradiation of GCE with ultrasound or activation using a carbon arc were proposed [5]. Engstrom and Strasser [14] performed such treatment by applying 1.75 V vs. SCE for 5 min followed by −1.0 V vs. SCE for 1 min. Kepley and Bard [15] also activated GCE by sweeping the potential to a high anodic value, and assumed that the graphite oxide layer formed was responsible for the activation. The low sensitivity obtained with unmodified GCEs caused that the surface is often modified with metals, organic substance, polymers or other modifiers [16]. However, modifications of carbon surfaces

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Table 1
Results of comparison of figures of merit of the present work with other studies.

Method	LOD/linearity [mg L^{-1}]	References
Spectrofluorometric	0.01/0.1–100.0	Vilchez et al. [32]
Capillary electrophoresis	0.038/0.076–15.12	Bohnenstengel et al. [33]
Ion-chromatography	0.060/0.060–0.120	Pérez and Bello [34]
Liquid chromatography	0.5/1.0–50.0	Monser and Darghouth [23]
Fluorescence polarization	0.99/0.99–199.5	Pettersson et al. [35]
Differential pulse voltammetry	0.027/0.030–226.8	Goyal et al. [36]
Capillary electrophoresis–chemiluminescence	0.0001/0.0001–0.010	Zhao et al. [37]
Efficient microemulsion liquid chromatography	0.2/4.0–20.0	McEvoy et al. [38]
HPLC–MS/MS	0.00000036/0–180	Hewavitharana et al. [39]
Square-wave adsorptive stripping voltammetry	0.0015/0.0015–0.30 and 0.30–3.00	Kachoosangi et al. [28]
Cyclic voltammetry	0.005/0.015–3.0	Kang et al. [40]
Adsorptive stripping voltammetry	0.0015/0.0076–10.6	Zhang et al. [41]
Differential pulse voltammetry	0.032/0.15–15.0	Fan et al. [42]
Normal pulse voltammetry + CWT	0.004/0.004–0.22	Present work

require additional reagents, and complex scan procedures have to be used in order to get an active and stable surface. Other challenges related to carbon electrodes are how to make the modified surface of the electrode reproducible and thus the response stable, how to avoid adsorption of organic substances and formation of passive films especially in real samples. Also CPEs have a complicated preparation process of regeneration of the surface and a short self-life [17].

In this work, an early study of the renewable glassy carbon annular band electrode (R-GC-ABE) as a suitable alternative of mercury-free voltammetric sensor, is presented. The possibilities of renovation of the solid electrodes in the device with the inside electrochemical cell were presented by us earlier [18]. The R-GC-ABE has a simple construction, it is easy to made and uncomplicated in usage. The idea of R-GC-ABE construction is derived from renewable silver ring electrode (RSRE) [19,20].

Paracetamol (PAR) [N-acetyl-p-aminophenol, acetaminophen] is an acylated aromatic amide, which was firstly applied in medicine as an antipyretic/analgesic specimen by Von Mering in 1893. Till now it has been in use as an effective analgesic and a non-steroidal inflammatory drug [21–23]. It is thought to be a very safe drug if recommended doses are not exceeded. This drug can also be used even in children and neonates [24,25]. Paracetamol is frequently prescribed solely or with other related drugs, but its mode of action is still not fully understood [22,26]. Therefore, determination of PAR in pharmaceuticals is of paramount importance, since an overdose of paracetamol can cause fulminating hepatic necrosis and other toxic effects [27]. Many methods have been reported for detecting PAR (Table 1). Consequently, the development and application of electrochemical sensors for the determination of PAR has received considerable interest in the past few decades, such as multiwalled carbon nanotube modified basal plane pyrolytic graphite electrode [28] or C_{60}-modified glassy carbon electrode [29].

Voltammetry is a group of highly sensitive analytical methods. One of them is normal pulse voltammetry (NPV). In this method the potential which is applied to the electrode, consists of a series of pulses of increasing amplitude, with the potential returning to the initial value after each pulse. The current is measured after generation of each potential pulse, after waiting time necessary for the charging current to decay. The sigmoidal-shaped curve is typically observed for NPV. The defect of NPV is poor resolution between neighboring waves because of drawn-out sigmoidal *I–E* response [30]. Therefore NPV technique is not often used in electrochemical measurements.

Signal processing may be considered as a one of the most powerful technologies that has great influence for science and engineering development in the twenty-first century. Signal processing procedures transform recorded curves what enables the extraction of relevant information. A mathematical formalisms describing time–frequency relations, which is recently often applied, is wavelet transformation (WT). The wavelet schemes involve projecting the time-domain signal into a wavelet domain where the base functions are wavelets, i.e. scaled and translated copies of a finite-length oscillating waveform known as a mother wavelet. The wavelets enable analysis by decomposing the signal into components of the different frequencies with a resolution that matches its scale [31]. Generally, WT enables the effective transformation of the sigmoidal wave-shaped, such as NPV, curves to the peak-shaped signals. The procedure proposed in this work utilizes the continuous wavelet transform (CWT) [30]. Application of the especially constructed mother wavelet which is the dedicated wavelet allows improvement of analytical parameters as: linearity, sensitivity, accuracy, detection limit, precision and others.

According to our knowledge, determination of paracetamol by NPV technique using the R-GC-ABE electrode has not been described earlier. Table 1 shows some advantages of present work compared with previous studies [32–42].

2. Theory

The described advanced numerical algorithm which utilizes wavelets, plays an important role in the signal processing in presented strategy of PAR determination. In electrochemistry wavelet transformations (WT) were often applied as tools which enables, for example, de-noising, baseline correction, separation of the overlapping signals. The CWT of the input signal $x(t)$ is expressed as:

$$CWT_x(a,b) = \int_{-\infty}^{\infty} x(t)\Psi_{ab}^{*}(t)\,dt \tag{1}$$

where the asterisk in (1) means the complex conjugation while, $\Psi_{ab}(t)$ can be derived from the mother wavelet $\Psi(t)$ by scaling and shifting according to the following formula:

$$\Psi_{ab}(t) = \frac{1}{\left|\sqrt{a}\right|}\Psi\left(\frac{t-b}{a}\right) \tag{2}$$

The result of the CWT are wavelet coefficients $CWT_x(a,b)$ which are the functions of a and b. The a and b are called scale and translation parameters, respectively. The coefficient $1/a$ may be interpreted as a measure of frequency. The parameter b, on the other hand, indicates the location of the wavelet window along the time axis.

Typically, in CWT mother wavelets were used, which are accessible in numerical software. In our work we did not use one of those previously defined in the literature but the especially constructed mother wavelet [30], calculated on the background of the ideal wave-shaped NPV curve. The dedicated mother wavelet

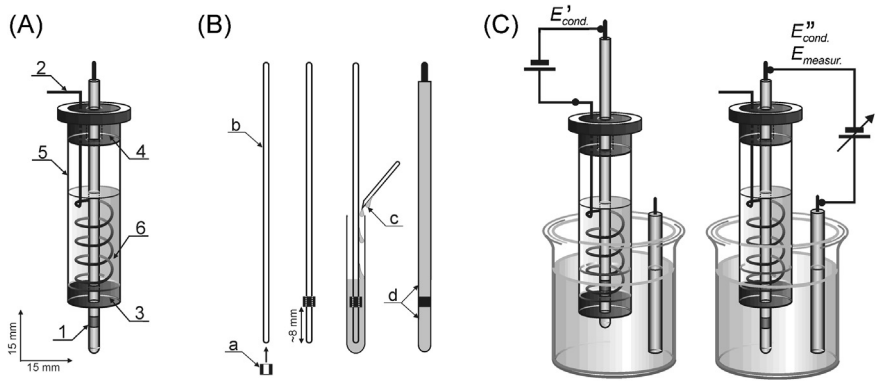


Fig. 1. (A) The R-GC-ABE used in all experiments: (1) glassy carbon annular band electrode (GC-ABE), (2) silver wire counter/quasi-reference electrode (AgQRE), (3) O-ring, (4) Teflon® centering element, (5) electrode body, (6) internal electrolyte. (B) Construction of the GC-ABE: (a) GC tube, (b) stainless steel wire, (c) and (d) resin. (C) The principle of electrochemical activation of the GC-ABE.

applied in calculation in the case of NP voltammetry curves allows transformation of inconvenient for the experimenter wave-shaped curves to the peak-shaped. The different components of the recorded curve may be separately observed by application of various scales (parameter a in (1) and (2)). Selection of an appropriate scale not only retains the important information of original data but also suppresses noise information. By usage of lower scales higher frequencies may be observed, while by higher scales, lower frequencies, which are often interpreted as an useful component of the signal. In our work different scales were applied and the optimal scale was manually selected by observation of the shape of the transformed curve. After preprocessing with CWT, the coefficients obtained were used in further regression analysis.

Calculation were realized using MATLAB system (The MathWorks Inc.), version R2009b, with Wavelet Toolbox version 4.4.1.

3. Experimental

3.1. Instrumentation

Experiments were carried out with the use of a electrochemical analyzer M161 (*mtm-anko*, Poland). All measurements were performed employing a classical three-electrode quartz voltammetric cell of volume 5 mL. As a working electrode a homemade renewable glassy carbon annular band electrode (R-GC-ABE) with the geometrical surface area of 0.18 cm^2, as a reference electrode Ag/AgCl/3 M KCl and as an auxiliary electrode Pt wire were applied. All potentials given are with respect to the potential of the reference electrode. Prior to use, glassware and, if necessary, also electrode body were cleaned by immersion in a 1:10 aqueous solution of HNO_3, followed by copious rinsing in distilled water. All data were obtained at room temperature.

3.2. Reagents and solutions

All the reagents of analytical grade, from Merck or Sigma chemical companies, were used without further purification. Double distilled water was used to prepare all solutions throughout the study. A fresh standard PAR solutions (1000 mg L^{-1}) was prepared by dissolving adequate amount of PAR in double distilled water. The supporting electrolyte was 0.02 mol L^{-1} potassium hydrogen phosphate (KH_2PO_4) and 0.01 mol L^{-1} perchloric acid ($HClO_4$), pH 2.2. Certified reference material: Seronorm™ Trace Elements Urine was obtained from the SERO AS Norway. The drug tablets which were used in the experiments were purchased from a local supplier. These were: APAP® Extra 500 mg tablets (500 mg Paracetamolum + 65 mg Coffeinum), APAP® Night tablets (500 mg Paracetamolum + 25 mg Diphenhydramini hydrochloridum) and Paracetamol (150 mg/5 mL) syrup (all products of US Pharmacia, Poland). The dimensions of the glassy carbon tube (Mineral, Poland) were: $\phi_{o.d}$ = 3 mm, $\phi_{i.d}$ = 1 mm diameter and 2 mm length. The resin was TRANSLUX D180 (AKSON, France).

3.3. NP and DP voltammetric procedures

The solution of the supporting electrolyte containing 0.02 mol L^{-1} KH_2PO_4 and 0.01 mol L^{-1} $HClO_4$ (pH 2.2) was prepared by mixing the appropriate volume of each stock solution in an electrochemical cell for total volume of 5 mL. Quantitative measurements were then performed using the differential and normal pulse mode (DP and NP). When an initial blank was recorded, the required volumes of the compound were added by means of a micropipette. In the present study, the optimal results for NPV and DPV experiments were obtained by a references potential (E_0) of 0.3 V (NPV), a resting period of 5 s, a step potential of 1 mV, a pulse potential (ΔE) of 30 mV (DPV) and a pulse width (t_p) of 40 ms (20 ms waiting time + 20 ms sampling current time).

3.4. The construction of the R-GC-ABE and the principle of electrode renovation

Fig. 1A shows the construction of the homemade R-GC-ABE electrode used in the experiments. The main parts of the R-GC-ABE are: glassy carbon annular band electrode (1); silver wire (2) used as a silver counter/quasi-reference electrode (AgQRE); an O-ring (3) ($\phi_{i.d}$ = 3 mm) made from 3 mm thick silicon rubber; Teflon® fastening element (4), fixed in the polypropylene body (5) filled with the supporting electrolyte spiked by 0.01 mol L^{-1} chloride ions (6).

Fig. 1B shows preparation of the glassy carbon annular band electrode (GC-ABE). The GC tube (a) was slid on a stainless steel wire (b) (ϕ = 1 mm) and fastened with conducting glue. The steel wire below and above the GC tube was covered with resins (c). The excess of the resin was then mechanically removed (d). After mounting, the electrode surface (resins and GC) was grinded with emery paper of decreasing grain size and polished with alumina powder (Buehler) of 1, 0.3 and 0.05 μm average particle size, and finally cleaned with methanol and distilled water in ultrasonic bath.

Fig. 1C clarifies the principles of operation of the R-GC-ABE. The single renovation and measurement cycle involves the following steps: 1) transferring the GC-ABE from the measuring cell containing the tested solution to the electrolyte inside the electrode body, 2) applying a constant or variable activation potential to the GC-ABE (vs. AgQRE), 3) shutting off the activation potential, immersing the GC-ABE in the measuring cell and beginning of

the measurements. During the insertion of the GC-ABE through the O-ring, the solid contaminants and bubbles of the gas (air, hydrogen) are removed from its surface. As an internal electrolyte an arbitrary solution may be used, but for a stable work of AgQRE it should contain min. 5 mmol L^{-1} of chloride ions. The total time of electrode regeneration lasts not longer than 10 s.

A R-GC-ABE electrode used in synthetic solutions, free of surface-active species (SAS) and organic matter, could by used over 100–200 stripping runs without electrochemical renovation and ca. one thousand stripping runs without mechanical polishing. However, an R-GC-ABE electrode used in the presence of the high concentration of SAS may require cyclic renovation before each measurement. In this case also internal electrolyte must be changed every 50–100 stripping runs. When the precision is worse than 5% the GC-ABE electrode must be repolished with alumina powder. The R-GC-ABE electrode while not in use was stored in distilled water.

3.5. APAP® and paracetamol syrup analysis

APAP® tablets (US Pharmacia, Poland), each containing 500 mg of paracetamol, were powdered and amounts corresponding to 1000 mg L^{-1} of PAR were weighted and dissolved in water. After sonication, working solutions were prepared by serial dilution. 4.950 mL of supporting electrolyte was placed in the voltammetric cell. After recording of the initial blank, 0.05 mL of APAP® solution (C_{PAR} = 100 mg L^{-1}) or 0.05 mL of Paracetamol syrup (diluted of double distilled water in the relation 1:300) was added by means of a micropipette and a voltammogram was recorded for the sample. In all the experiments, voltammograms were recorded under the same conditions as for pure PAR.

3.6. Analysis of spiked urine samples

Spiked urine solutions were prepared as follows: 0.050 and 0.100 mL of PAR stock solution (1000 mg L^{-1}) were placed in 1 mL flasks and filled up to volume with human urine (CRM). Voltammograms of the blank were recorded after placing 0.100 mL of pure urine and 4.900 mL of supporting electrolyte in voltammetric cell. Subsequently, 0.100 mL of the spiked urine solution and 4.900 mL supporting electrolyte were placed in voltammetric cell and voltammogram was recorded for the sample.

4. Results and discussion

In this work experiments which relied on the determinations of the very low levels of paracetamol concentration at the R-GC-ABE, were performed by DPV and NPV techniques parallel to demonstrate the advantages of the new approach, in which the important step was signal processing.

4.1. Preliminary experiments at R-GC-ABE electrode

4.1.1. Electrochemical characteristic of the R-GC-ABE electrode

The working potential window of the R-GC-ABE electrode in the used herein supporting electrolyte was −0.9 to 1.5 V. For the R-GC-ABE electrode before activation no or very low peak/wave of PAR was observed, while in case of the activated electrode a single peak/wave is well formatted. For PAR concentrations below 0.1 mg L^{-1} the peak (for DPV) and below 5 μg L^{-1} the wave (for NPV) height increases linearly with the increase of PAR concentration. The electrochemical activation of R-GC-ABE electrode can be performed in non-deaerated supporting electrolyte in three ways. The first one consists in multiple polarization of the GC-ABE electrode at two different potentials −1.2 and 1.4 V, applied alternately every 5–10 s. However, this method is not efficient enough, it can hardly be controlled and was used only for re-activation of the electrode. The second way is performed by periodic, manifold polarization of the electrode with linearly increasing potential in the range −1.2 to 1.4 V. That method of metallic electrode activation is also recommended in the literature [1,2,4,5]. The parallel recording of the cyclic *I–E* curves makes it possible to control the progression of the electrode renovation. But satisfactory reproducibility of parameters of the activated electrode was achieved after at least ca. 20–30 cycles of polarization. In the third method, the R-GC-ABE electrode is at first polarized at the constant potential −1.2 V for 30 s, followed by a full cycle of polarization from −1.2 to 1.4 V. 2–3 cycles of activation followed by alumina polishing, were sufficient to regain the electrode stability.

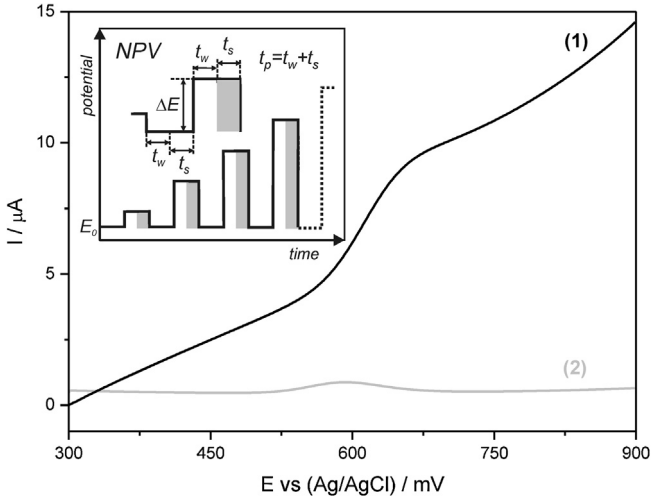


Fig. 2. NPV (1) and DPV (2) curves of 5 mg L^{-1} paracetamol at the R-GC-ABE in supporting electrolyte: 0.02 M KH_2PO_4, 0.01 M $HClO_4$. DP and NP mode: potential step, 1 mV; pulse width, 40 ms; pulse amplitude, 30 mV (DP); reference potential, 0.3 V (NP). Inset – a normal pulse potential–time pattern.

4.1.2. Re-activating the R-GC-ABE electrode

To ensure repeatability of the registered voltammograms and measurements, precision higher than 2% is inevitable to re-activate electrochemically the GC-ABE electrode. Thus, each of measurements cycle consisted of two-step electrode activation – initially for 5–10 s at a potential $E'_{cond.} = -1.2$ V in internal electrolyte (inside the electrode body) and later for 2–3 s at potential $E''_{cond.} = 0.3$ V in the measuring cell containing the tested solution (Fig. 1C).

4.2. Signal processing of the paracetamol NP voltammograms

By application of NPV sigmoidal (wave-shaped) curves were recorded. Fig. 2 presents comparison of voltammograms recorded for the same paracetamol concentration by DPV and NPV techniques, which instrumental parameters were optimized. A normal pulse potential–time pattern is presented in inset in Fig. 2. Measurements were carried with supporting electrolyte spiked with 5 mg L^{-1} of PAR. It was observed that the sensitivity of NP voltammograms is ca. 32 times greater than DP recordings. The half-wave potential of PAR NP curves is moved to negative values by ca. 20 mV in relation to the peak position in DP voltammograms.

Wave-shaped curves are often difficult in further interpretation despite of problems with precise determination of the wave height. Therefore we propose application of the advanced signal processing algorithm based on wavelet theory. In our experiments recorded curves were transformed to the wavelet coefficients space by CWT and dedicated mother wavelet. In the effect of such transformation peak-shaped curves were obtained. In calculations we

applied continuous transformation because it does not decrease signal resolution.

After choice of the adequate mother wavelet also scale – the second most important parameter in CWT should be indicated. In each case by selection of the optimal scale the useful component of the signal may be extracted. It was observed that low values of the scale gives asymmetric signal and which was difficult in further interpretation because of the low height of the obtained peak. Also, it is a known fact that lowest level of the scale does not allow to improve signal-to-noise ratio by removal of random noise. On the other hand, usage of the relatively great scales results in deterioration of the signal, it is unable to determine precisely peak position.

In the process of the manual selection of the optimal scale we observed that the interval of the scale 120–180 provides peak-shaped symmetric curves with the bottom parallel to the *x*-axis. The proposed signal processing procedure gives the possibility to avoid the interpretation of wave-shaped curves and moreover the relevant parameters of the registered data used in the quantitative chemical analysis are not disturbed. It holds true when the same scale was applied in transformation of the whole calibration set.

4.3. Analytical utility of R-GC-ABE electrode

4.3.1. Optimization of the parameters of the DP and NP voltammetry

In order to adopt the DP and NP voltammetry methods to trace concentrations of paracetamol, electrolytes and instrumental parameters were optimized.

At the beginning, signals of PAR in various types and concentrations of electrolytes such as: phosphate buffer (pH 6.3), acetate buffer (pH 4.5), potassium chloride, potassium nitrate, perchloric acid, potassium hydrogen phosphate, etc. were measured (Fig. 3A). High sensitivity of PAR determination was obtained in KCl solution, the lowest and simultaneously the simplest for interpretation in respect to background current was obtained in $HClO_4$. But in KCl environment the precision was unsatisfactory. Best results (precision, reproducibility and near linear shape of the baseline before and after the wave) were obtained for the last two solutions (potassium hydrogen phosphate and perchloric acid), so wide range of concentrations for each of them was tested. In the effect, the solution of the supporting electrolyte containing 0.02 mol L^{-1} KH_2PO_4 and 0.01 mol L^{-1} $HClO_4$ (pH 2.2) was chosen. Additionally, in pH near 2, the effectiveness of interferences removal from the electrode surface was most satisfactory as a result of hydrogen evolution reaction.

The optimized parameters were as follows: accumulation potential (E_{acc}) and time (t_{acc}), step potential (E_s), pulse amplitude (ΔE) for DPV, pulse width (t_p) and reference potential (E_0) for NPV. Consequently, these parameters were investigated with supporting electrolyte spiked with 5 mg L^{-1} PAR. Fig. 3 presents the results of optimization for NPV technique. The values were calculated as a heights of the peaks obtained in CWT transformation of the NP sigmoidal curves.

4.3.1.1. Effect of accumulation potential and time. The influence of the accumulation potential was studied in the range from 0.1 to 0.5 V for DPV and NPV. In this potential range for NPV technique the magnitude of the analytical signals (slightly) decreased linearly with the increase of accumulation potential (ca. 6%/0.1 V). The effect is not connected with the electrode process, but only with the nonlinear shape of the bottom of the NPV wave in the range from 0.1 to 0.5 V. In the case of the DPV technique, changes of the peak current height with preconcentration time were not observed. For E_{acc} = 0.30 V the proposed algorithm for transformation of the NPV sigmoidal-shaped signals guaranteed the best precision of the wave height (I_w) readings. Therefore the registration of DP and NP voltammograms begun from 0.30 V.

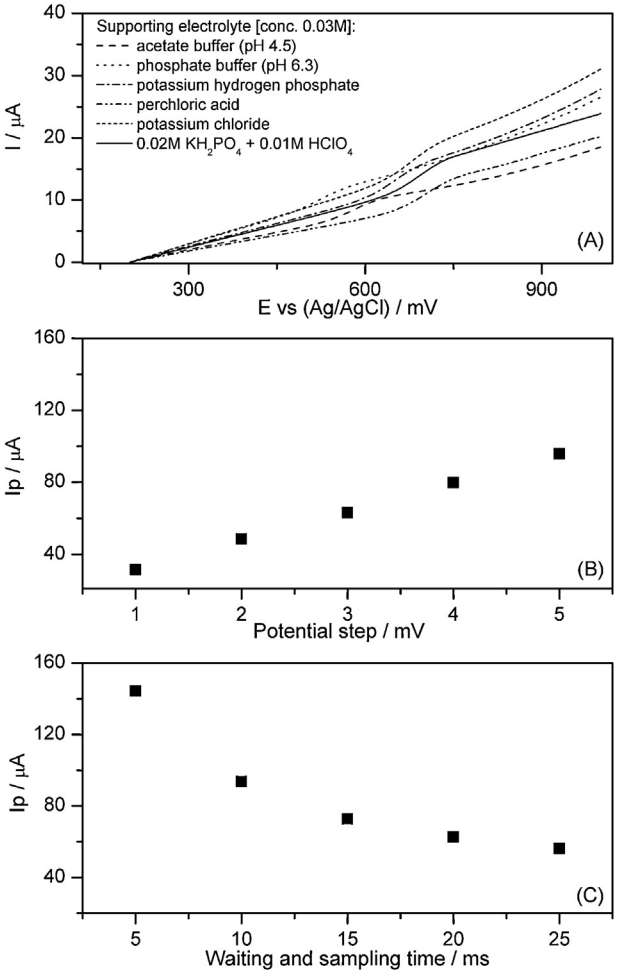


Fig. 3. (A) NP voltammograms of 5 mg L^{-1} paracetamol at the R-GC-ABE in differential supporting electrolyte: (type of electrolyte is marked on the plot). The influence of the paracetamol peak current (peak heights after CWT transformation in scale 150) corresponding to changing: (B) potential step, (C) pulse width (waiting time + sampling current time).

The dependence of wave and peak current versus the accumulation time was tested in the time interval from 0 to 30 s. In this range the wave height (I_w) and peak current (I_p) did not change and their values remained constant during the accumulation time change. Therefore, the stage of accumulation was neglected.

4.3.1.2. Optimizing the remaining parameters. The other optimized parameters are: potential step, pulse width and pulse amplitude for DPV. The values of these parameters influenced both, the value of current and shape of the curves. In order to determine the appropriate values of parameters the measurements were made in a wide range for each of them.

At the beginning, the various values of starting (E_p) and reference (E_0) potentials (for NPV) were tested in the range from 0.0 to 0.5 V. It was observed that they have no influence on the signal. Therefore $E_p = E_0 = E_{acc}$ were chosen for further experiments.

The wave height increased with the increase of potential step (Fig. 3B). It may be explained by the increase of the working electrode polarization rate. The diffusion layer grows slower, therefore local PAR concentration near the R-GC-ABE surface is higher and as a result also PAR oxidation current increases. The optimal value of the potential step was chosen as 1 mV, selected from the range

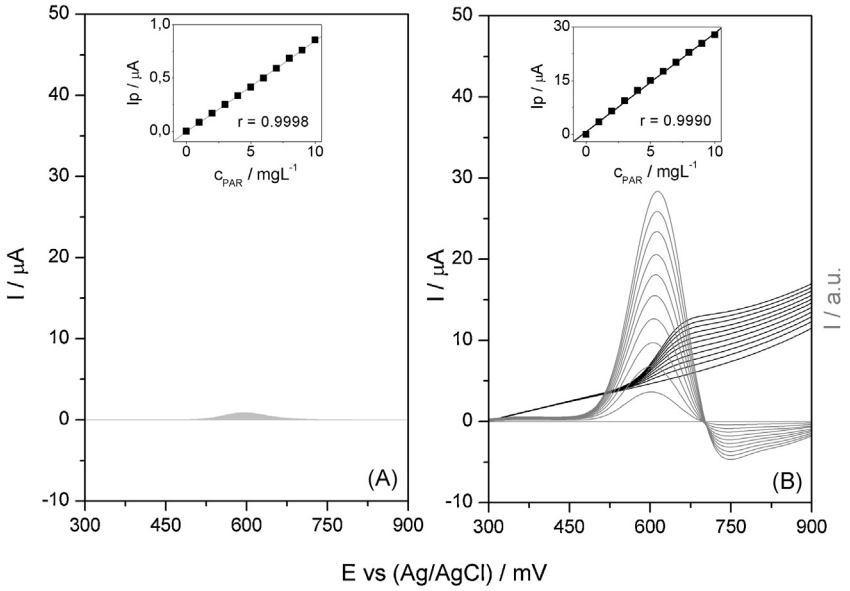


Fig. 4. The calibration plots for paracetamol at the R-GC-ABE in supporting electrolyte. Concentration of PAR: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10 mg L^{-1}. (A) DPV curves, (B) NPV curves – original and after CWT transformation in scale 150. Inset – calibration plots. Other conditions as in Fig. 2.

1–5 mV. Application of the potential steps higher than 1 mV caused substantial increase of the background current.

The pulse width (t_p) was changed from 10 to 50 ms. In each case it was assumed that $t_w = t_s$ (waiting time = current sampling time). The PAR wave height decreased with the lengthening of the pulse width (Fig. 3C), what may be explained by Cottrell equation, because the recorded current deceases according to electrolyze time as $\sim t^{-1/2}$. The best result (precision, reproducibility and the signal-to-background current ratio) was obtained for $t_p = 40$ ms, and this was the value chosen for further work.

4.3.2. Sensitivity, linearity range, detection limit

To verify the linear relationship between anodic peak/wave currents and PAR concentrations several calibration curves were constructed under optimal conditions and supporting electrolyte. Fig. 4A and B shows differential (DP) and normal (NP) pulse voltammograms and corresponding calibration curves obtained for blank and PAR solutions of concentration from 1 to 10 mg L^{-1}. Additionally, Fig. 4B demonstrates curves before and after CWT transformation done by optimal scale. The interpretation of the obtained peak-shaped curves are easier than experimental sigmoidal curves. The calculated parameters of calibration curve, and detection limit are presented in Table 2. The most important improvement which was obtained by application of NPV was more than 7–8 times increase of the experimental sensitivity [4]. The slope of the calibration curve after additional application of signal processing procedure realized with the usage of CWT was in the effect 32 times higher than in the case of DPV (Table 2). Other analytical parameters are comparable and fulfill typical validation criteria.

The voltammetric curves of PAR in both cases were obtained by optimal parameters therefore the comparison is reliable. Finally, it may be concluded that the applied signal processing strategy is useful in paracetamol determinations and in the effect gives satisfactory values of analytical parameters of the method.

4.3.3. Interferences

Voltammetric signals are strongly affected by the type and concentration of all the components of measured electrolyte. The effects of dissolved oxygen, ions and different organic compounds were investigated for the R-GC-ABE electrode.

One of the advantages of GCEs is the fact, that the results of voltammetric measurements are only little influenced by the presence of oxygen in the analyzed solution. Therefore, all the measurements could be performed in solutions without initial oxygen removal.

Surfactants distort the analytical signal (peak and wave height) – during the measurements surfactants and organic matter adsorption takes place on the working electrode. Therefore, continuous renovation of the electrode, or at least its surface, is necessary. It is especially easy in the case of the hanging mercury drop electrode (HMDE) and the R-GC-ABE. The successive generated mercury drops (HMDE) and the successive renovation cycles for the R-GC-ABE definitively solves the problem of electrode renovation, even before each measurement.

Determination of PAR generally suffers from the interference of ascorbic acid, tyrosin, D-glucose, L-glutamic acid, dopamine and uric acid. The influences of common interfering species were investigated in the presence 5 mg L^{-1} PAR. The results showed that concentrations of ascorbic acids (50), tyrosin (40), D-glucose (500), L-glutamic acid (200), dopamine (20), uric acids (15) did not change the signal of NP voltammograms, where the data in the brackets are concentration ratios of the interfering species to that of the analyte.

Table 2
Comparison of the calibration parameters estimated for technique DPV and NPV after CWT transformation.

Technique	$a \pm sa$ [µA/mg L^{-1}]	$b \pm sb$ [µA]	r	DL [mg L^{-1}]
DPV	0.0854 ± 0.0005	−0.006 ± 0.003	0.9998	0.02
NPV + CWT	2.75 ± 0.04	0.9 ± 0.2	0.9990	0.004

a: sensitivity; b: intercept.

Table 3
Results of determination of PAR content in two commercial tablets (APAP® Extra: Paracetamol, Caffeine; APAP® Night: Paracetamol, Diphenhydramine hydrochloride) and syrup (Paracetamol) using R-GC-ABE electrode by NPV technique and ultraviolet method.

Preparation	Declared amount [mg]	Found [mg]	CV [%]	Recovery[b] [%]
APAP® Extra	500	457 ± 10[a]	11.7	92
APAP® Night	500	486 ± 3[a]	3.7	97
Paracetamol syrup[c]	3.00	3.15 ± 0.19[a]	0.8	105

[a] $t(S/n^{1/2})$, $p = 95\%$, $n = 5$.
[b] Recovery = $[(C_{anal+addition} - C_{anal})/C_{addition}] \times 100\%$.
[c] mg mL^{-1}.

Table 4
Results of the PAR determination in spiked urine by NPV technique, $n = 5$.

Volunteer	Added [mg L^{-1}]	Found [mg L^{-1}]	CV [%]	Recovery[b] [%]
1	1.00	1.09 ± 0.08[a]	5.3	109
2	2.00	2.11 ± 0.13[a]	3.9	106

[a] $t(S/n^{1/2})$, $p = 95\%$, $n = 5$.
[b] Recovery = $[(C_{anal+addition} - C_{anal})/C_{addition}] \times 100\%$.

4.3.4. Stability and reproducibility of the R-GC-ABE

The stability of the R-GC-ABE electrode was checked by recording successive DP and NP voltammograms for supporting electrolyte. After 100 cycles, no change was observed in the voltammetric profiles of the R-GC-ABE. Using the electrode to detect PAR by normal pulse voltammetry in supporting electrolyte containing 2 mg L^{-1} PAR, slight change (1–2%) was observed in 20 successive detections. Using it to detect 5 different PAR solutions (same concentration), the relative standard deviation observed was 2.6%. Furthermore, when the R-GC-ABE was stored in distilled water, no significant change in the response was observed for more than two weeks. Over longer periods (two weeks), the sensitivity of the signal fluctuates by about 5%. These experiments indicated that the R-GC-ABE electrode has good stability and repeatability for the determination of PAR.

4.3.5. Analysis of APAP® and paracetamol syrup

One tablet of APAP® Extra contains 500 mg PAR and 65 mg caffeine. One tablet of APAP® Night contains 500 mg PAR and 25 mg diphenhydramine hydrochloride. Peak deriving from caffeine is visible at a potential ca. 1.5 V and does not interfere with PAR determination. Diphenhydramine hydrochloride is not electrochemically active in considered potential range. PAR concentration was analyzed using the standard addition method. Five replicate experiments were performed with additions as described in Section 3. The recovery results for PAR in tablets and syrup are given in Table 3.

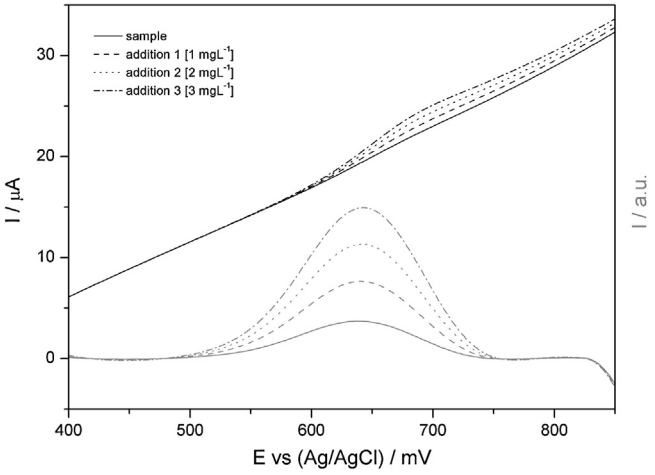


Fig. 5. NP voltammograms of PAR determination in spiked human urine (spiked amount – 1 mg) using standard addition method; additions indicated by each line. Experimental conditions the same as in Fig. 2.

4.3.6. Analysis of spiked urine samples

The standard addition method was used to determine PAR in spiked urine samples. Three equal additions of standard were made, as described in Section 3. Urine was analyzed using the standard addition method. Typical voltammograms obtained during experiments are shown in Fig. 5. The recovery results for PAR in spiked urine are given in Table 4.

5. Conclusions

In this work we introduced the new working electrode – renewable glassy carbon annular band electrode (R-GC-ABE). The results of application of R-GC-ABE and NPV technique for determination of paracetamol (PAR) exhibited very good sensitivity and selectivity. Additionally it offers LOD comparable with AdSV technique and even lower than CV and DPV methods. The proposed in this work strategy is simpler and does not require such expensive apparatus as liquid chromatography or capillary electrophoresis methods. The feature of the proposed strategy is application of the signal processing algorithm which gives the possibility of conversion of the wave-shaped curves into the peak-shaped which are more friendly in further interpretation. The peak-position on the *x*-axis of the output curve is the same as position of the inflection point, what means that the important feature of the analytical signal does not change. The relation between the waves heights is also preserved, therefore the calibration could be made in the wavelet coefficients space. Finally it should be underlined that the proposed strategy is a useful tool which resolves the problem of the processing of sigmoidal voltammetric signals recorded in the determination of PAR in synthetic and real samples. The effects of potential interfering ions were studied, and it was found that the proposed procedure is free from interferences of most common interfering ions and organic compounds. The proposed electrode was used in determination of PAR in some real samples like diluted commercial tablets, syrup and human urine (CRM), without the necessity for sample pretreatment or any time-consuming extraction or evaporation steps prior to the analysis, with satisfactory recovery. The simple renovation procedure, reproducibility, high stability, wide linear dynamic range, low detection limit and high sensitivity suggest that this electrode is an attractive candidate for practical applications.

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